Machine Learning for Bioinformatics

Exercise Sheet

Freie Universität Berlin

Please note that the jupyter notebook must be submitted along with the exercise sheet![1]

Name:		Matriculation no.:	
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Model selection

Assignment 1. *PRESS for linear and ridge regression*

1. Complete the implementation of *MyRidgeRegressor*. Points: 0
2. What is the minimum and maximum value of the PRESS statistic for varying p? (Format: N.NN) Points: 1

   Minimum PRESS: Maximum PRESS:

3. What is the minimum and maximum value of LOO-CV with mean squared error for varying p? (Format: N.NN) Points: 1

   Minimum LOO-CV: Maximum LOO-CV:

Assignment 2. *Degrees of freedom (DF)*

1. Complete the implementation of the function *df*. Points: 0
2. What is the degree of freedom for $\alpha = 0.5$? (Format: N.NN) Points: 1

   DF:

3. What is the degree of freedom for $p = 1, 2, \ldots, 50$? Points: 1

   n: p: $\max(n, p)$: $\min(n, p)$:

[1] No points are awarded if an answer is only partially correct. Answers must be supported by results from the jupyter notebook.

Assignment 3. *Regularization paths*

1. At the beginning of the ℓ_1 regularization path all features have coefficients equal to zero. Which is the first feature (1-based column index in F) with non-zero coefficient? Points: 1

   Feature:

2. Which of the following statements holds for the ℓ_2 regularization path? Points: 1

   The sum of absolute values of the coefficients increases with α.

   Correct:

   Some of the coefficients change sign.

   Correct:

   The curves are smoother than for the ℓ_1 path.

   Correct:

   Similar to the ℓ_1 path, there are α-regions of non-zero length where coefficients return back to zero.

   Correct:

Assignment 4. *Implicit regularization*

1. Complete the implementation of the function *compute_noisy_polynomial_features*. Points: 0

2. For which noise level(s) (scale / standard deviation) do you observe the minimum average mse after the interpolation threshold? Points: 2

   0.01: 0.02: 0.05:

Assignment 5. *Theory questions*

1. The BIC is used to approximate the marginal likelihood $p(X \mid m_i)$, where m_i is a statistical model. Assume we have two models, m_1 and m_2, which both fit the data X equally well. However, m_2 has more parameters than m_1. The BIC approximated marginal likelihood for m_1 is therefore Points: 1

   higher: lower:

   than the marginal likelihood of m_2.

2. The Bayesian Information Criterion (BIC)

   $$\text{BIC} = -2 \log L(\hat{\theta}) + p \log n$$

   assumes that model complexity increases monotonically with the number of parameters p. In modern over-parameterized models the test error can follow a double-descent curve. Which of the following statements are correct? Points: 2

Beyond the interpolation threshold, test error can decrease even as p increases.
In this regime, the BIC penalty $p \log n$ can over-penalize large models.
BIC always tracks the effective complexity of over-parameterized models.

Maximum number of points: 11